A manager looks at last month’s sales report and asks three questions in one meeting: “What happened? Why did it happen? What should we do next?” Each question needs a different kind of analysis.

CHAPTER 2 — THE DATA ECOSYSTEM

Types of Analytics

Lesson 1 of 6

BY THE END

You can tell descriptive, diagnostic, predictive, and prescriptive questions apart.

What is analytics?

Analytics means using data to understand what happened, why it happened, what may happen next, or what action to take.

PLATE I — THE FOUR TYPES OF ANALYTICS

Descriptive

PLATE I — THE FOUR TYPES OF ANALYTICS

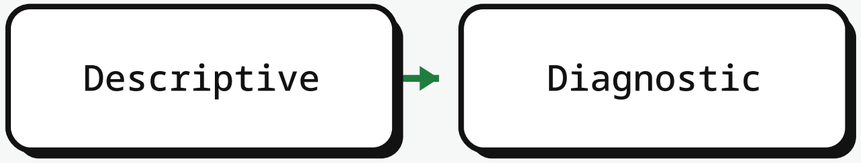

PLATE I — THE FOUR TYPES OF ANALYTICS

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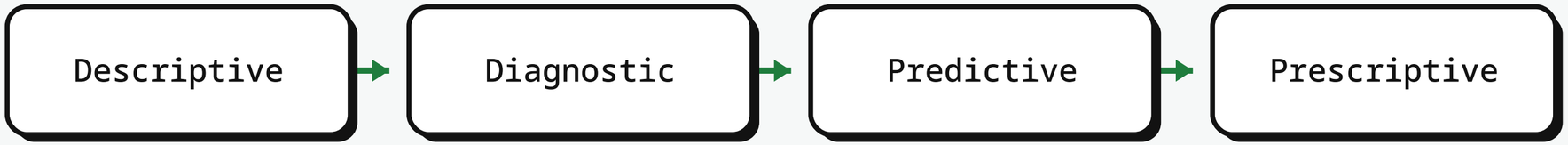

each question needs the one before it answered

The four types of analytics

TYPE	QUESTION	EXAMPLE
Descriptive	What happened?	Rice sales rose 12% last month
Diagnostic	Why did it happen?	They rose because the Hargeisa shop stayed open through Ramadan
Predictive	What might happen next?	Sales may stay high next month if the longer hours continue
Prescriptive	What should we do?	Keep the longer hours in Hargeisa, and try them in Muqdisho

Simple way to remember them

- **Descriptive** = report the past
- **Diagnostic** = explain the past
- **Predictive** = forecast the future
- **Prescriptive** = recommend action

Most beginners start with descriptive analytics, such as summaries, totals, and charts. More advanced work adds prediction and recommendation.

The four are cumulative

You cannot explain why rice sales rose until you know that they rose. You cannot forecast next month without knowing what drove last month. And a recommendation made without the three steps underneath it is just an opinion with a number attached to it.

4 — WHY THIS MATTERS

So when an organization says it wants prediction, the honest first question is whether it can yet describe what already happened.

KEY TERMS

TERM	DEFINITION
Analytics	The use of data to find patterns and support decisions
Descriptive analytics	Summarizes what already happened
Diagnostic analytics	Explains why something happened
Predictive analytics	Estimates what may happen in the future
Prescriptive analytics	Recommends actions based on analysis